

LAVENDER CALHOUN

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EDUCATION

Wayne State University

Sep 2021 - May 2024

Bachelor's, Computer Science

GPA: 3.68

- Cum Laude, 4x Dean's List Student, Society of Computer Developers Member, 6x Vocal Music Grant Awardee

PROFESSIONAL EXPERIENCE

Argonne National Laboratory

Lemont, IL, USA

SULI Intern

May 2024 - Aug 2024

- Contributed to a Python library for streamlining machine learning workflows on Linux systems with high-performance computing nodes.
- Automated dataset generation for molecular dynamics simulations, reducing manual processing by 68%.
- Optimized training cycles by creating workflows to generate datasets addressing model weaknesses.

Wayne State University

Detroit, MI, USA

Undergraduate Student Researcher

Dec 2022 - Apr 2023

- Explored the Neural Radiance Fields (NeRF) algorithm, focusing on its architecture, principles, and applications.
- Refined data preprocessing and model optimization, reducing novel view interpolation error by 15%.

Wilson Adaptive Technologies

Detroit, MI, USA

Frontend Design Intern

Jun 2022 - Aug 2022

- Created mockups and redesigned for four website pages to enhance user experience, accessibility for customer needs, and branding.
- Developed a comprehensive style guide; standardizing color palettes, fonts, and design elements following the redesign.
- Conducted three user interviews and created detailed profiles to inform design decisions.

MTU Great Lakes Research Center

Houghton, MI, USA

CADeT Cyber Initiative Student Employee

Sep 2020 - Apr 2021

- Developed hands-on experience in machine learning by using the YOLO v3 algorithm to identify armored fighting vehicles from photographic datasets.
- Processed, analyzed, and visualized 5,000+ energy usage data points to identify seasonal trends using Excel, as well as the Pandas, Matplotlib and Seaborn Python libraries.
- Led a presentation on research findings, demonstrating the practical applications of machine learning and data analysis techniques.

PROJECTS & OUTSIDE EXPERIENCE

Tune Tracer

Detroit, MI, USA

Team Lead/ML Lead

Jan 2024 - Apr 2024

- As Team Leader, directed the agile development of a cross-platform app according to client needs and project risks.
- Applied a machine learning model to predict the modality of a song with 94% accuracy.
- As a full-stack developer, designed a record/upload audio function, UI elements in Figma and Dart, and trained the machine learning model.
- Created an API server to handle prediction requests using Flask and Heroku.

SKILLS

Skills: Git, Python, HTML/CSS, Java, SQL, Tensorflow, Pytorch, Jupyter, NumPy, Pandas, Heroku, MongoDB, Flask, CSS, HTML, React.js, Bash, Canva, C/C++, Computer Vision, Dart, Data Science, Docker, Figma, Flutter, Firebase, JavaScript, Keras, Linux/Unix, Machine Learning, MySQL, Seaborn, Software Testing, UI/UX Design, Web Development, Word/Pages/Docs, Data Structures & Algorithms, Node.js, Data Analysis, Project Management, Problem Solving, Communication